Asiful Arefeen

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Summary

– Biomedical Data Science PhD candidate at Arizona State University with a Master's in Computer Science. Research focused on diabetes, metabolic health, and AI-driven clinical decision support systems.
– Published in leading biomedical informatics and digital health venues including ACM HEALTH, JAMIA Open, IEEE JBHI, EMBC, BHI, BSN, and CHASE.
– Recipient of the **NIH T32 Institutional Training Grant** for AI in Precision Nutrition (AIPrN).
– 5+ years of research experience in diabetes and digital health using machine learning, wearable sensing, multimodal physiological data, and longitudinal glucose monitoring datasets.
– Awarded $55,000 in research funding as Champion of the 2024 College of Health Solutions Heat and Health Research Challenge.
– Expertise in glucose forecasting, hyper/hypoglycemia prevention, explainable AI, personalized behavioral intervention modeling, precision nutrition, multimodal data analysis, and AI-driven treatment recommendation systems for individuals with Type 1 Diabetes.
– Collaborated with Mayo Clinic endocrinologists to conduct user studies and analyze longitudinal CGM and insulin pump data for development of clinically actionable diabetes intervention strategies.
– Experienced in interdisciplinary collaboration with endocrinologists, nutrition scientists, data scientists, and engineers to develop translational digital health solutions.
– **U.S. citizen with no visa sponsorship requirements; open to relocation anywhere in the U.S.**

Education

Arizona State University — Aug 2021 - Sept 2026

PhD in Biomedical Informatics & Data Science — *Aug 2021 - Sept 2026*
MS in Computer Science — *Jan 2024 - Dec 2025*
MS in Biomedical Informatics — *Aug 2021 - May 2023*

- **Advisor:** Dr. Hassan Ghasemzadeh – website: ghasemzadeh.com
- **PhD Thesis:** Toward AI-Driven Interventions for Improved Glucose Outcomes
- **MS Thesis:** Label-Efficient Artificial Intelligence in Digital Health
- **Notable Courses:** Machine Learning, Neural Network Design, Embedded Machine Learning, Data Mining, Linear Algebra, Clinical & Health Informatics, Foundations of Biomedical Informatics Methods

Bangladesh University of Engineering & Technology — Feb 2015 - Apr 2019

BS in Electrical & Electronic Engineering

- **Thesis:** Measuring and assessing the effects of harmonics on the power grid during EV charging
- Published three papers in IEEE conference proceedings

Research Interests

- **AI for Diabetes and Metabolic Health:** AI-driven diabetes and metabolic health research focused on improving glucose outcomes through personalized behavioral and clinical interventions.
- **Explainable and Trustworthy AI:** Developing interpretable and clinically trustworthy AI models that provide transparent treatment recommendations and decision-support tools for endocrinologists and patients.
- **Personalized Behavioral Interventions:** Modeling personalized behavioral interventions using longitudinal CGM, insulin pump, dietary, and wearable sensor data to support prevention of hyperglycemia and hypoglycemia.
- **Digital Health and Mobile Health (mHealth):** Leveraging smartphones, wearable sensors, and multimodal physiological data for remote monitoring, patient engagement, and chronic disease management.

- **Precision Nutrition and Dietary Assessment:** Applying machine learning and multimodal sensing to study metabolic responses, dietary behaviors, and personalized nutrition recommendations.
- **Computational Modeling and Digital Twins:** Building computational models and digital twins to simulate individualized responses to diabetes interventions and lifestyle modifications.
- **Generative and Label-Efficient AI:** Developing generative AI, active learning, and self-supervised learning approaches for healthcare applications with limited labeled clinical data.
- **Clinical Decision Support Systems:** Integrating machine learning, longitudinal health data, and patient-specific modeling to develop clinically actionable decision-support systems for translational diabetes research.
- **Clinical Data Extraction and Visualization:** Transforming structured and unstructured healthcare data into actionable insights through data extraction, integration, and visualization methods.

Experience

Idaho National Laboratories, Idaho | *AI, Computer Vision Intern* May 2026 – Aug 2026
– I will be developing AI methods to convert plant drawings/schematics into a list/graph of connected equipment.

Mayo Clinic (Endocrinology Dept.), Arizona | *Research Affiliate* Oct 2023 – April 2026
– Developed *GlyMan*, a counterfactual intervention technique to assist current automated insulin delivery (AID) technology in diabetes management. GlyMan surpasses baselines (DiCE, NICE, CFNOW) in validity and proximity. Working on *GlucoGuide*, an end-to-end counterfactual explanations software to assist endocrinologists in clinical decision making.

Embedded Machine Intelligence Lab, ASU | *Grad Research Assistant* Aug 2021 – Present
– Developed novel counterfactual technique *ExAct* for behavioral intervention towards better diabetes outcome. Performance wise, ExAct improves validity by at least 10% and proximity by at least 6.6% compared to baselines like DICE, NICE, CEML, and Optbinning.

Washington State University | *Grad Teaching & Research Assistant* Aug 2020 – July 2021
– Performed research on motion artifact suppression in physiological signals. TA for Computer Security, Data Structures C/C++, Program Design and Development. Held lab sessions and office hours. Set and graded quizzes, graded assignments.

Awards & Achievements

Fellowship: 1. NIH T32 Institutional Training Grant for AI in Precision Nutrition (AIPrN) Research (2024-ongoing). 2. ASU Graduate College Completion Fellowship 2026.

Research Grant: Champion of 2024 College of Health Solutions (CHS) Student Heat and Health Research Challenge Award. Awarded $55,000 in research funding.

Best Poster: 1. Best poster award at the 2026 ASU College of Health Solutions (CHS) Faculty Research Day.
2. 2nd Runner up Demo-Poster at the 025 IEEE-EMBS International Conference on Wearable and Implantable Body Sensor Networks (BSN'25), Los Angeles, CA, USA.
3. Best poster award at the 2024 and 2026 ASU College of Health Solutions (CHS) Faculty Research Day.

Travel Award: NSF Student Travel Award to attend IEEE BHI 2024, BHI 2023, ACM CHASE 2022, ASU GSG Travel Award for EMBC 2025 in Copenhagen, Denmark, 2x ASU CHS Travel Support in 2025 for EMBC 2025 and BSN 2025, IEEE BHI 2025 NSF-EMBS-Google Sponsored Young Professional NextGen Scholar

Grant: ASU Graduate College University Grant 2022-23, 2023-24

Teaching Experience

1. **Instructor,** BMI 310: App Development for Population Health, Arizona State university, Spring 2024.
2. **Guest Lecturer,** Wearable Sensing and Machine Learning for Precision Nutrition, BMI 570: BMI Seminar Series, Arizona State university, Spring 2023.

Invited Talks

1. Towards Generating User-Centric and Realistic Counterfactuals for Positive Health Outcomes, Truveta, Seattle, WA, July 2025.

2. Towards AI-Driven Behavioral Interventions in Digital Twin Systems for Improved Health Outcomes, Control Systems Engineering Laboratory (CSEL), Tempe, AZ, June 2025.

Workshops

1. AI/ML for Digital Health Interventions in the 2025 Emerging Digital Methods to Enable Precision Health series held at Phoenix Bioscience Core, Arizona State University (November 7th, 2025).

Publications

Peer reviewed Journals

1 **Arefeen, A.**, Khamesian, S., Grando, M. A., Thompson, B., & Ghasemzadeh, H. (in press). GlyTwin: Enhancing digital twin for glucose control in type 1 diabetes using patient-centric counterfactual treatments. IEEE Journal of Biomedical and Health Informatics.

2 **Arefeen, A.**, Singh, S.J., Razavi, C., Ghasemzadeh, H., & Dev, S. (2025). Assessing the quality of reporting in artificial intelligence/machine learning research for cardiac amyloidosis. JAMIA open, 8 5, ooaf104.

3 Mamun, A., **Arefeen, A.**, Racette, S.B., Sears, D.D., Whisner, C.M., Buman, M.P., & Ghasemzadeh, H. (2025). LLM-Powered Prediction of Hyperglycemia and Discovery of Behavioral Treatment Pathways from Wearables and Diet. *Sensors (Basel, Switzerland)*, 25.

4 **Arefeen, A.**, & Ghasemzadeh, H. (2025). Cost-Effective Multitask Active Learning in Wearable Sensor Systems. *Sensors (Basel, Switzerland)*, 25.

5 **Arefeen, A.**, Akbari, A., Mirzadeh, S., Jafari, R., Shirazi, B., & Ghasemzadeh, H. (2023). Inter-Beat Interval Estimation with Tiramisu Model: A Novel Approach with Reduced Error. *ACM Transactions on Computing for Healthcare*.

6 Mirzadeh, S., **Arefeen, A.**, Ardo, J., Fallahzadeh, R., Minor, B.D., Lee, J., Hildebrand, J.A., Cook, D., Ghasemzadeh, H., & Evangelista, L.S. (2022). Use of machine learning to predict medication adherence in individuals at risk for atherosclerotic cardiovascular disease. *Smart health, 26*.

Peer reviewed Conferences

7 Soumma, S.B., **Arefeen, A.**, Carpenter, S.M., Hingle, M., & Ghasemzadeh, H. (2025). SenseCF: LLM-Prompted Counterfactuals for Intervention and Sensor Data Augmentation. In Proceedings of the *IEEE-EMBS International Conference on Body Sensor Networks (BSN'25).*

8 Khamesian, S., **Arefeen, A.**, Thompson, B., Grando, M.A., & Ghasemzadeh, H. (2025). AZT1D: A Real-World Dataset for Type 1 Diabetes. In Proceedings of the *IEEE-EMBS International Conference on Body Sensor Networks (BSN'25).*

9 **Arefeen, A.**, Fessler, S.N., Mostafavi, S.M., Johnston, C., & Ghasemzadeh, H. (2025). MealMeter: Using Multimodal Sensing and Machine Learning for Automatically Estimating Nutrition Intake. In Proceedings of the *47th Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC'25).*

10 **Arefeen, A.**, & Ghasemzadeh, H. (2025). LEAD: Localized explanations with adversarial decision boundary characterization for interpretable disease prediction. In Proceedings of the *47th Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC'25).*

11 **Arefeen, A.**, Khamesian, S., Grando, M.A., Thompson, B., & Ghasemzadeh, H. (2024). GlyMan: Glycemic Management using Patient-Centric Counterfactuals. *2024 IEEE EMBS International Conference on Biomedical and Health Informatics (BHI)*, 1-5.

12 Shroff, P., **Arefeen, A.**, & Ghasemzadeh, H. (2023). GlucoseAssist: Personalized Blood Glucose Level Predictions and Early Dysglycemia Detection. *2023 IEEE 19th International Conference on Body Sensor Networks (BSN)*, 1-4.

13 **Arefeen, A.**, & Ghasemzadeh, H. (2023). GlySim: Modeling and simulating glycemic response for behavioral lifestyle interventions. *2023 IEEE EMBS International Conference on Biomedical and Health Informatics (BHI)*, 1-5.

14 **Arefeen, A.**, Fessler, S.N., Johnston, C., & Ghasemzadeh, H. (2022). Forewarning Postprandial Hyperglycemia with Interpretations using Machine Learning. *2022 IEEE-EMBS International Conference on Wearable and Implantable Body Sensor Networks (BSN)*, 1-4.

15 **Arefeen, A.**, Jaribi, N., Mortazavi, B.J., & Ghasemzadeh, H. (2022). Computational Framework for Sequential Diet Recommendation: Integrating Linear Optimization and Clinical Domain Knowledge. *2022 IEEE/ACM Conference on Connected Health: Applications, Systems and Engineering Technologies (CHASE)*, 91-98.

16 Alinia, P., Parvaneh, S., Mirzadeh, S., **Arefeen, A.**, & Ghasemzadeh, H. (2022). Boosting Lying Posture Classification with Transfer Learning. *44th Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC'22)*, 109-114.

17 **Arefeen, A.**, Shehreen, S., Islam, A.K., Rahman, M., & Rahman, H. (2019). Measuring and Assessing the Effects of Harmonics on the Grid of Bangladesh Power System During EV Charging. *2019 IEEE International Conference on Power, Electrical, and Electronics and Industrial Applications (PEEIACON)*, 5-9.

Preprints

18 Soumma, S.B., **Arefeen, A.**, Carpenter, S.M., Hingle, M., & Ghasemzadeh, H. (2026). Counterfactual Modeling with Fine-Tuned LLMs for Health Intervention Design and Sensor Data Augmentation. ArXiv, abs/2601.14590.

19 **Arefeen, A.**, Soumma, S.B., & Ghasemzadeh, H. (2025). RealAC: A Domain-Agnostic Framework for Realistic and Actionable Counterfactual Explanations. *ArXiv, arXiv:2508.10455*.

20 Khamesian, S., **Arefeen, A.**, Grando, M.A., Thompson, B., & Ghasemzadeh, H. (2025). Type 1 Diabetes Management using GLIMMER: Glucose Level Indicator Model with Modified Error Rate. *ArXiv, abs/2502.14183*.

21 **Arefeen, A.**, & Ghasemzadeh, H. (2023). Designing User-Centric Behavioral Interventions to Prevent Dysglycemia with Novel Counterfactual Explanations. *ArXiv, abs/2310.01684*.

Conference Presentations

Oral

1 LEAD: Localized Explanations with Adversarial Decision Boundary Characterization for Interpretable Disease Prediction at the 47th Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC'25), Copenhagen, Denmark.

2 GlySim: Modeling and Simulating Glycemic Response for Behavioral Lifestyle Interventions at the 2023 IEEE EMBS International Conference on Biomedical and Health Informatics (BHI'23), Pittsburgh, PA, USA.

3 Computational Framework for Sequential Diet Recommendation: Integrating Linear Optimization and Clinical Domain Knowledge at the 2022 IEEE/ACM Conference on Connected Health: Applications, Systems and Engineering Technologies (CHASE'22), Washington, DC, USA.

Poster

4 MealMeter: Using Multimodal Sensing and Machine Learning for Automatically Estimating Nutrition Intake. at the 47th Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC'25), Copenhagen, Denmark.

5 GlyMan: Glycemic Management using Patient-Centric Counterfactuals at the 2024 IEEE EMBS International Conference on Biomedical and Health Informatics (BHI'24), Houston, TX, USA.

6 GlucoGuide: Retrospective Counterfactual Decision Support to Address Stakeholder Needs at the 2025 IEEE EMBS International Conference on Biomedical and Health Informatics (BHI'25), Atlanta, GA, USA.

Demo

7 GlucoGuide: A Retrospective Counterfactual Decision Support to Address Stakeholder Needs at the 2025 IEEE-EMBS International Conference on Wearable and Implantable Body Sensor Networks (BSN'25), Los Angeles, CA, USA.

Professional Services

- Journal reviewer for ACM Transactions on Computing for Healthcare (HEALTH)
- Journal reviewer for PLOS Digital Health
- Journal reviewer for IEEE Journal Of Biomedical & Health Informatics (JBHI)
- Journal reviewer for ACM Transactions on Intelligent Systems and Technology (TIST)
- Journal reviewer for Nature Portfoloio Scientific Reports
- Journal reviewer for Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies
- Reviewer for International Conferences of the IEEE Engineering in Medicine and Biology Society (EMBC 2025)
- Reviewer for International Conferences of the IEEE-EMBS International Conference on Biomedical and Health Informatics (BHI 2023, 2024, 2025)
- Reviewer for International Conferences of the IEEE International Conference on Pervasive Computing and Communications (PerCom 2023)
- Reviewer for International World Wide Web Conference (WWW 2026)

Mentorship Experience

Research Mentor, Arizona State University: One Master's student and one High-School student

1. Aashritha Machiraju (MS, CS, 2024-2025), in thesis formulation and conference publication.
2. Prisha Shroff (Hamilton High School, Chandler, AZ, 2023, current affiliation: undergrad student in Computer Science at Stanford), mentorship led to publishing a research paper on glycemic management using AI (GlucoAssist in IEEE BSN 2023).

Ongoing Projects

GlyTwin: Digital Twin for Glucose Control in Type 1 Diabetes Through Behavior Change | *GitHub*

- We enhance the capabilities of traditional digital twin paradigm with counterfactual interventions/explanations. We are working on getting it validated with physicians and users.

FoodOCR: Information Extraction from Handwritten Food Logs | *GitHub*

- In this project, we had several participants who logged their meals on paper. Later, we used Google Cloud Vision OCR to extract the handwritten food logs. As the extractions were not 100% accurate, the extracted texts were fixed using BERT through prompt engineering.

GlySynth: Autoregressive Framework for Synthetic Glycemic Response Generation | *GitHub*

- A conditional autoregressive scheme for generating synthetic physiological signals (preferably glycemic response) given contextual information. As validated by downstream classification task, GlySynth produces better signals compared to GANs under data scarcity.

MealMeter: Meal Macronutrient Estimation with Multimodal Sensing and Machine Learning | *GitHub*

- MealMeter is a lightweight Machine Learning framework for nutrient intake tracking, trained and validated on controlled data collected from 12 healthy individuals, improves carb and fat estimation errors by at least 10.8% and 21.9%, respectively, compared to the baselines (multitask-learning and self-supervised learning).

MetaPlate: Counterfactual-Guided RAG-LLM for Food Recommendation and Anomaly Prevention | *GitHub*

- MetaPlate is a personalized nutrition recommendation system that forecasts postprandial blood glucose responses from glucose, wearable sensors, and logged meal data. It integrates counterfactual modeling with LLM to translate metabolic optimization into food-level suggestions that minimizes glucose spikes while preserving user preferences.